

80386 Operating Modes (Real, Protected & Virtual 8086) -- MPCA Module 5

Definition: The 80386 microprocessor supports three distinct operating modes: Real Address Mode (16-bit 8086 compatible), Protected Virtual Address Mode (32-bit multitasking protected environment), and Virtual 8086 (V86) Mode (running real-mode 8086 software inside protected mode).

1. Detailed Comparison of 80386 Operating Modes

80386 OPERATING MODES		
1. REAL MODE:	2. PROTECTED MODE:	3. VIRTUAL 8086:
- Default upon power-on	- PE bit = 1 in CR0	- VM bit = 1
- Exact 8086 clone	- 32-bit Flat/Segmented	- Run 8086 apps
- 1 MB Physical RAM	- 4 GB RAM / 64 TB Virtual	as task under
- No memory protection	- Hardware Privilege 0-3	multitasking OS

2. Deep Dive into Each Mode

1. Real Address Mode (Real Mode)

- * Processor boots in this mode upon power-up/reset.

2. Protected Virtual Address Mode (Protected Mode)

- * Activated by setting PE = 1 in Register CR0.

3. Virtual 8086 Mode (V86 Mode)

- * Sub-mode of Protected Mode enabled by setting VM = 1 in EFLAGS register.

3. Operating Mode Comparison Table

Property	Real Mode	Protected Mode	Virtual 8086 Mode
Address Space	1 MB	**4 GB Physical / 64 TB Virtual**	1 MB per V86 task
Memory Protection	None	**Full 4-Level Rings (0-3)**	Protected under Ring 3
Multitasking	Not supported in hardware	**Hardware Task Switching (TSS)**	Multitasked by OS kernel
Paging Support	Disabled	**Full 4 KB Paging (PG bit)**	Supported via Paging MMU

4. Quick Recall Flow

Power-on -> Real Mode (8086 clone 1MB) -> Set PE=1 in CR0 -> Protected Mode (4GB, Rings 0-3) -> Set VM=1 -> Virtual 8086 Mode